



Performing solvation free energy calculations in LAMMPS using the decoupling approach

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Abstract

The decoupling approach to solvation free energy calculations requires scaling the interactions between the solute and the solution with all intramolecular interactions preserved. This paper reports a new procedure that makes it possible to these calculations in LAMMPS. The procedure is tested against built-in GROMACS capabilities. The model compounds chosen to test our methodology are ethanol and biphenyl. The LAMMPS and GROMACS results obtained are in good agreement with each other. This work should help perform solvation free energy calculations in LAMMPS and/or other molecular dynamics software having no built-in functions to implement the decoupling approach.

Keywords Solvation free energies · Thermodynamic integration · Molecular dynamics · LAMMPS

Introduction

Solvation free energy is the free energy change associated with transferring a solute molecule from an ideal gas phase to a solution phase at a given temperature, pressure, and solute concentration in solution. Solvation free energy computations play an important role in computing solubilities [1–8], partition coefficients [9–11], activity coefficients [12] and Henry's law constants [13]. Established free energy perturbation (FEP) [14] and thermodynamic integration (TI) [15] methods for computing the solvation free energy fully account for solute–solvent interactions in all solute and solvent arrangements via different alchemical pathways. One approach, called decoupling, modulates only the interactions between the solute and its surrounding, retaining internal interactions. Several molecular dynamics (MD) codes (GROMACS [16], AMBER [17], etc.) include tools to implement the decoupling approach but a key step of charge scaling without altering the intramolecular coulombic interactions cannot be implemented in LAMMPS [18]. This note demonstrates a procedure to implement the decoupling

approach in LAMMPS without altering the source code (an example of altering the source code can be found in the work of Paluch et al. [19] where they modify the code of the MD simulation package MDynaMix [20, 21]). Beyond LAMMPS, our procedure can be implemented with any MD code with equivalent functions as described in this note.

Thermodynamic cycle

In order to perform a free energy calculation we need to define the end states. For solvation free energy calculations one end state (as shown in Fig. 1) is the isolated molecule (A) in the gas phase along with the solution phase (B) in the background, i.e., state (1). The other end state is the solvated molecule in the solution (B) phase, i.e., state (6). In state (1), the atoms of the isolated molecule interact only with each other. Since the molecule being solvated is isolated from the surrounding, we refer to this state as the decoupled state, because the atoms are not interacting with (decoupled from) the surrounding environment. In state (6), the atoms of the molecule interact with all the atoms present in the system, including the solvent and other dissolved solutes. We refer to this state as the coupled state. The thermodynamic cycle shown in Fig. 1 depicts the alternate paths between the end states for solvating a molecule [22]. We can construct numerous alternate paths for computing solvation free energies, the

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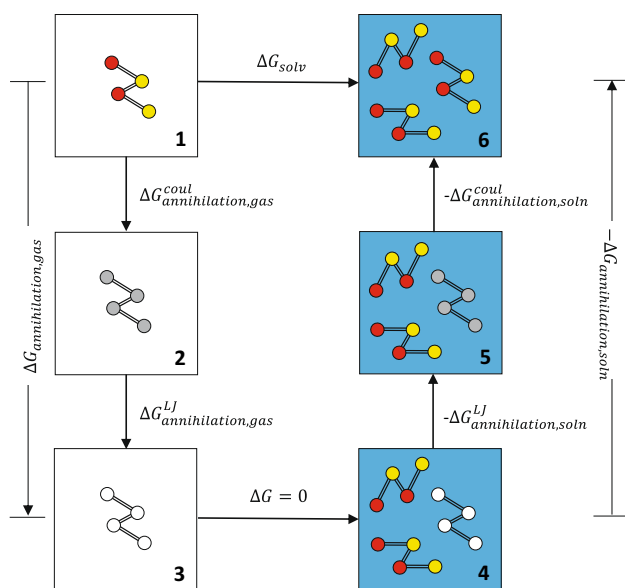


Fig. 1 The Thermodynamic Solvation Energy Cycle [22], white box: isolated gas phase; blue box: water; colored atoms: LJ=ON, coul=ON, grey atoms: LJ=ON, COUL=OFF, and white atoms: LJ=OFF, coul=OFF. The interactions of the solute under discussion are inter+intra for solution phase and intra only for the isolated gas phase. In state (1), (2) and (3) the solution phase (into which the isolated molecule is being solvated) is present in the background (as explicitly shown in Fig. 2) to complete the reactant side of the physical reaction of solvation. For clarity, we do not show the solution phase on the left side of the figure

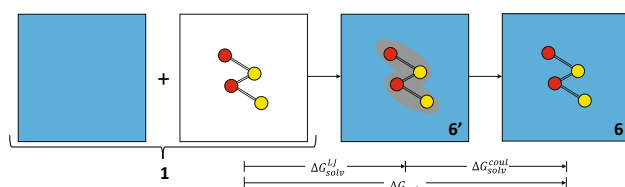


Fig. 2 The decoupling thermodynamic path. Grey shaded area: LJ_{A-B} = ON, COUL_{A-B} = OFF; colored atoms: intramolecular LJ and coulombic interactions = ON

two common implemented paths are: (i) The annihilation route, (1)→(2)→(3)→(4)→(5)→(6) and (ii) The **decoupling route**, (1)→(6) with an intermediate state (6') as shown in Fig. 2. The annihilation route involves annihilation of the solute's pairwise (inter + intra) interactions in the gas and solution phases. The thermodynamic path (1)→(2)→(3) is the gas phase annihilation path, where the coulombic interactions are turned off from (1)→(2) followed by turning off of the Lennard–Jones (LJ) interactions from (2)→(3), in the isolated gas phase. The thermodynamic path (6)→(5)→(4) is the solution phase annihilation path, where the coulombic interactions are turned off from (6)→(5) followed by turning off of the LJ interactions from (5)→

(4), in the solution phase. The decoupling route involves decoupling the solute from its surrounding environment by turning off the solute–solution pairwise interactions.

In order to compute the free energy difference between states (1) and (6), we need to sum the free energy differences between successive states along the chosen route. The annihilation route can be implemented in LAMMPS but so far the decoupling route cannot as scaling the charges on the solute to decouple it from the solution does not preserve its intramolecular interactions. However, we have developed a methodology using which the decoupling route can be implemented in LAMMPS. In order to compute the free energy change along path (6)→(1) we need to compute the free energy change of turning off the solute–solution (A–B) interactions leaving the intramolecular solute interactions turned on.

Setting up the decoupling route

The free energy change along the path (1)→(6')→(6) is determined by computing the free energy change of turning on the LJ interactions between the solute (A) and the solution (B), followed by the free energy change of turning on the coulombic interactions between them. We use thermodynamic integration (TI) [15] to compute these free energy changes, where

$$\Delta G_{(i) \rightarrow (j)} = \int_{\lambda=0}^{\lambda=1} \left\langle \frac{dU}{d\lambda} \right\rangle_{\lambda} d\lambda \quad (1)$$

and $U(\lambda)$ is a Hamiltonian that continuously changes from that of system (i) to system (j) as λ goes from 0 to 1. The decoupling route is shown in Fig. 2. Here we are performing a hydration free energy calculation as the molecule is being solvated in pure water. Throughout states (1), (6') and (6) the solute's intramolecular LJ and coulombic interactions are on. State (1) comprises the isolated gas phase molecule and the water box. In state (6') only the solute–water (intermolecular) LJ interactions are on, and in state (6) the solute–water LJ and coulombic interactions are on. We use Gauss–Legendre quadrature to select intermediate λ 's from (1) to (6') and from (6') to (6). The free energy changes computed along these paths, ΔG_{solv}^{LJ} and ΔG_{solv}^{coul} , respectively, are then added to give the solvation free energy (ΔG_{solv}).

To compute ΔG_{solv}^{LJ} , the LJ interactions are turned on via the soft-core potential [23]. A coupling parameter (λ_{LJ}) is varied between 0 and 1 to march from state (1)→(6'), where $\lambda_{LJ} = 0$ means there are no LJ interactions between A and B, and $\lambda_{LJ} = 1$ means the LJ interactions are completely turned on. Likewise, for the electrostatics, a coupling parameter (λ_{coul}) is varied between 0 and 1 to compute ΔG_{solv}^{coul} , where

$$\Delta G_{solv}^i = \int_{\lambda_i=0}^{\lambda_i=1} \left\langle \frac{dU}{d\lambda} \right\rangle_{\lambda_i} d\lambda_i \quad (2)$$

where, i represents LJ or coul.

Staging the Lennard Jones interactions

Turning on the LJ interactions between A and B, i.e., coupling them, is straightforward in LAMMPS. The soft-core potential used in LAMMPS is given by:

$$U_{LJ}(r_{ij}; \lambda_{ij}) = \lambda_{ij}^2 4\epsilon \left\{ \frac{1}{\left[\frac{(1-\lambda_{ij})^2}{2} + \left(\frac{r_{ij}}{\sigma_{ij}} \right)^6 \right]^2} - \frac{1}{\frac{(1-\lambda_{ij})^2}{2} + \left(\frac{r_{ij}}{\sigma_{ij}} \right)^6} \right\} \quad (3)$$

In LAMMPS we can assign a λ_{LJ} value to each pairwise interaction between the atom types. Therefore, we set

$$\begin{aligned} \lambda_{ij} &= \lambda_{LJ} \quad \forall \quad i \in A, j \in B \\ &= 1 \quad \forall \quad i = j \in A \end{aligned} \quad (4)$$

This enables us to scale the A–B LJ interactions without altering the A–A LJ interactions. However, one important point to note while staging the LJ interactions is that we also need to keep the solute's intramolecular coulombic interactions turned on. We achieve this in our framework by setting the solute's charge scaling parameter (described in the following subsection), λ_q , equal to 10^{-9} . This is because in the overlay potential (also described in the following subsection), described by Eq. (7), we cannot insert integer zero as the value for λ_q . Hence, we set it to 10^{-9} (which is as an approximate to zero).

Staging the coulombic interactions

For turning on the coulombic interactions between A and B, we need to scale the charges on A in order to correctly account for the long range electrostatic interactions via the particle mesh Ewald sum. However, in doing so we scale the intramolecular coulombic interactions of A as well. This is not acceptable for the decoupling approach as we want the intramolecular interactions of A to be “unaltered”. Thus, what we want is

$$U_{coul}^{intra}(r_{ij}) = \sum_{i < j} k \frac{q_i q_j}{r_{ij}}; \quad \forall i, j \in A \quad (5)$$

where $k = 1/(4\pi\epsilon)$, ϵ being the permittivity of free space. However, since we scale $q_i \rightarrow \lambda_q q_i \forall i \in A$, the intramolecular (A–A) electrostatic interactions become,

$$\begin{aligned} U_{coul}^{intra, scaled}(r_{ij}; \lambda_q) &= \sum_{i < j} k \frac{(\lambda_q q_i)(\lambda_q q_j)}{r_{ij}} \\ &= \sum_{i < j} \lambda_q^2 k \frac{q_i q_j}{r_{ij}} \end{aligned} \quad (6)$$

The scaling of charges on the solute thus reduces the intramolecular interaction from that of an ideal gas. To recover the correct intramolecular Coulomb energy, we overlay,

$$\begin{aligned} U_{coul}^{overlay}(r_{ij}; \lambda_q) &= U_{coul}^{intra}(r_{ij}) - U_{coul}^{intra, scaled}(r_{ij}; \lambda_q) \\ &= \sum_{i < j} (1 - \lambda_q^2) \left\{ k \frac{q_i q_j}{r_{ij}} \right\} \\ &= \left(\frac{1 - \lambda_q^2}{\lambda_q^2} \right) U_{coul}^{intra, scaled}(r_{ij}; \lambda_q) \end{aligned} \quad (7)$$

over the $U_{coul}^{intra, scaled}(r_{ij}, \lambda_q)$ interactions, to recover $U_{coul}^{intra}(r_{ij})$. This overlay of potential is implemented in LAMMPS by using the *hybrid overlay* command and overlaying a *coul/cut/soft* potential. This potential has the following functional form (using the scaled charges)

$$U_{coul/cut/soft}(r_{ij}; \lambda_{coul}) = \sum_{i < j} \lambda_{coul} k \frac{(\lambda_q q_i)(\lambda_q q_j)}{r_{ij}} \quad (8)$$

where λ_{coul} is the user input parameter which we input as $(1 - \lambda_q)^2 / \lambda_q^2$ (see Eq. 7). This exact overlay correction is the main contribution in this work.

Testing

In order to test the procedure laid out for computing ΔG_{solv}^{coul} , we compute it for two compounds: (i) ethanol and (ii) biphenyl using LAMMPS and test the results against the results obtained from GROMACS which has a built-in function for performing solvation free energies using the decoupling approach. We compare $\langle dU/d\lambda \rangle_\lambda$ at each λ state outputted by LAMMPS and GROMACS, as well as the total coulombic free energy change to test our approach.

Simulation details: LAMMPS

We perform hydration free energy calculations using the procedure laid out in the above sections. Ethanol and biphenyl described by the General Amber Force Field (GAFF, version 1.7) [24] and AM1-BCC charges [25, 26] were solvated in 1288 TIP3P water molecules [27]. The Lorentz–Berthelot

mixing rules [28–30] were used for computing the interatomic pair coefficients. The initial configurations were taken from the FreeSolv database [31]. We used a time step of 1.0 fs. Bonds with hydrogen were constrained using SHAKE [32, 33]. Nonbonded interactions were cutoff at 1 nm, with long-range electrostatics handled by LAMMPS' Particle-Particle-Particle-Mesh (PPPM) summation [34, 35]. Long-range dispersion corrections to energy and pressure were applied. The NPT ensemble was simulated using Nose–Hoover thermostat and barostat [36] at $T = 298.15$ K and $P = 1.01325$ bar. All systems were equilibrated for 3 ns followed by 5 ns production runs. A 15 point Gauss–Legendre quadrature method [37] was used to evaluate the integral in Eq. (2) for computing the coulombic solvation free energy. For computing the solvation free energy due to the LJ interactions we used a 10, 15, and 15 point Gauss–Legendre quadrature for $0 < \lambda_{LJ} < 0.2$, $0.2 < \lambda_{LJ} < 0.4$, and $0.4 < \lambda_{LJ} < 1.0$, respectively, amounting to a total of 40 stages. Data every 1 ps was used to compute the thermodynamic averages.

Simulation details: GROMACS

For the GROMACS simulations, we follow procedures as identical as possible to those described in the FreeSolv database [31], only modifying run input files taken from the FreeSolv repository [38] minimally in order to ensure compatibility with the GPU-enabled GROMACS version 2016.1 [39]. Briefly, a neighbor list was updated every 10 steps with a cutoff for its construction of 1.2 nm. Nonbonded interactions were cutoff at 1.0 nm, with long-range electrostatics handled by particle mesh Ewald techniques [40] and a switching function applied for Lennard–Jones interactions between 0.9 and 1.0 nm. Long-range dispersion corrections to energy and pressure were applied, with all bonds involving hydrogens constrained via the LINCS algorithm [41] and water kept rigid via SETTLE [42]. At each of the λ states described in other sections, we performed 2500 steepest descent steps to minimize fully solvated structure and topology files downloaded from the FreeSolv database [31]. After minimization, a Langevin integrator [43] was used throughout to propagate dynamics with a 1 fs time step. We first equilibrated temperature at constant density for 25 ps, then simulated in the NPT ensemble for 25 ps to equilibrate density with both a tightly coupled Berendsen thermostat and isotropic barostat [44], and finally performed a 3.0 ns run in the NPT ensemble with a Langevin thermostat (stochastic integrator) [43] and isotropic Parinello–Rahman barostat [45, 46], which were also used for production. Production runs for 5.0 ns were performed at a constant temperature of 298.15 K and pressure of 1.01325 bar, with derivatives with respect to λ and potential energies at all states outputted every 1 ps to be averaged for thermodynamic integration (TI).

Table 1 Results for $\Delta G_{solv}^{coul}/k_B T$

Compound	LAMMPS	GROMACS	Difference
Ethanol	-8.73 ± 0.02	-8.76 ± 0.02	0.03 ± 0.03
Biphenyl	-8.30 ± 0.02	-8.29 ± 0.02	-0.01 ± 0.03

Results and discussion

We test our approach to compute the solvation free energies for turning on the coulombic interactions between the solute and solution, ΔG_{solv}^{coul} , in LAMMPS, by comparing our results against GROMACS' results (which uses *intmol* functionality to correctly account for the intramolecular interactions). The results are shown in Table 1 which are in good agreement. The error bars on the values represent a standard 68% confidence interval (C.I.). This means approximately 1 in every 3 simulations will give results outside the specified interval. Further, Fig. 3 compares the integrand for turning on the Coulomb interactions, $\langle dU/d\lambda_q \rangle_{\lambda_q}$ as a function of λ_q . As can be seen, the integrands are in good agreement too. Thus, we have demonstrated and tested our procedure for performing solvation free energy calculations in LAMMPS using the decoupling approach via counterbalancing the scaling of charges on the solute by overlaying a compensating potential to keep the intramolecular interactions unchanged to that of an ideal gas.

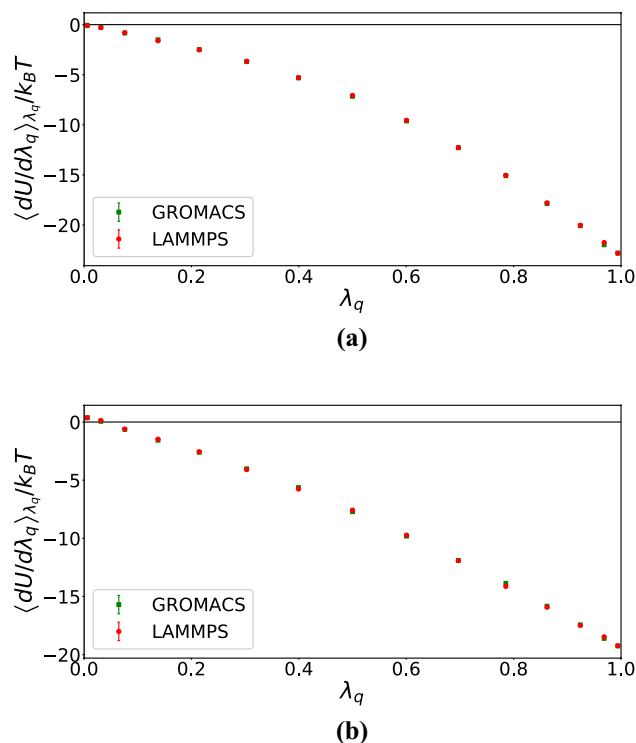


Fig. 3 $\langle dU/d\lambda_q \rangle_{\lambda_q}/k_B T$ as a function of λ_q for **a** ethanol, **b** biphenyl

Table 2 Results for $\Delta G_{\text{solv}}/k_B T$

Compound	LAMMPS	GROMACS	FreeSolv [31]
Ethanol	-5.76 ± 0.03	-5.79 ± 0.03	-5.72 ± 0.03
Biphenyl	-5.22 ± 0.04	-5.27 ± 0.04	-5.30 ± 0.05

We report the total (dimensionless) solvation free energies ($\Delta G_{\text{solv}}/k_B T$) in Table 2 for completeness. Our values are consistent across the two tests, and agree with the values computed by Mobley et al. [47] considering a 95% C.I. and that Mobley et al. used 15 and 5 stages, whereas we used 40 and 15 stages for turning on the LJ and coulombic interactions, respectively. Note: We use GAFF version 1.7 instead of 1.8 in order to make a comparison to the values reported in [31].

Conclusions

We have introduced and tested a procedure to carry out solvation free energy calculations in LAMMPS using the decoupling approach. This procedure can be implemented in any MD code which allows the user to define ‘overlay’ potentials. We have only compared solvation free energies between LAMMPS and GROMACS for two molecules to check the implementation because the overlay correction is, in principle, exact. Also, while we have chosen to implement thermodynamic integration here for simplicity, the overlay correction should work equally well with other free energy calculations such as Bennett Acceptance Ratio (BAR) [48] and Multistate Bennett Acceptance Ratio (MBAR) [49].

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